

Sven van Holten Charria

Pre-doctoral Research Assistant at the Centre for Macroeconomics, LSE

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EDUCATION

London School of Economics and Political Science 2024 – 2025

MSc Econometrics and Mathematical Economics (distinction)

Selected modules: Econometric theory, Further topics in econometrics, Advanced macroeconomics.

Erasmus University Rotterdam 2019 – 2024

Double BSc in Econometrics & Economics (honours)

Selected modules: Advanced econometrics, Macroeconometrics, Time-series econometrics, Analysis, Probability theory, Statistical theory, Stochastic processes, Seminar in forecasting.

Chinese University of Hong Kong 2023

Exchange semester, Department of Economics

Selected modules: Machine learning for economics, Chinese exchange rate reforms.

RESEARCH POSITIONS

Pre-doctoral Research Assistant 2025 – present

Profs. Ricardo Reis, Benjamin Moll & Jonathan Hazell, LSE

- Assisted in research on “*Heterogeneous-Agent Macro and Expectations*” (Moll), “*Parallel Currencies and Financial Repression*” (Reis), and “*Empirical Macro*” (Hazell).
- Selected contributions: built reinforcement-learning solvers for neoclassical growth models, solved two-period and infinite-horizon parallel-currency models analytically and computationally, constructed historical postwar transfer panels and HANK-ready microdata, and encoded UK micro-prices for ML inflation forecasting.

RESEARCH PAPERS

Perceived Income Risk and Its Dynamics Across the Income Distribution — *Working Paper*.

- **Topics:** Perceived income risk, diagnostic expectations, heterogeneous-agent models.
- **Findings:** Risk perceptions show persistent, income-graded dynamics absent from realised earnings; specifically, surprises at the bottom raise risk far more persistently than at the top.
- **Application:** Embedded in an Aiyagari economy with state-dependent diagnostic expectations, the resulting excess precautionary saving suppresses consumption even after income recovers.

Intuitive Inference: Enhancing Post-Double Selection Inference on Treatment Effects with Intuition-Based Penalties — *BSc Thesis*, supervised by Stan Koobs.

- **Topics:** High-dimensional inference, causal machine learning, Monte Carlo simulation.
- **Findings:** Belief-based penalties integrate economic intuition into the post-double Lasso selection step, reducing variance in treatment-effect inference while maintaining similar bias.
- **Application:** Re-examining Donohue & Levitt (2001) on abortion and crime, the method yields a smaller, economically defensible control set with tighter treatment-effect inference.

MRF-ARCH: A Machine Learning Approach to Forecast USD/GBP Tail Risk in a High Dimensional Dataset — *Seminar Paper*, supervised by Prof. Robin Lumsdaine.

- **Topics:** Time-series econometrics, volatility forecasting, random forests, tail risk.
- **Findings:** A derived ARCH(p)–AR(p) equivalence embeds time-varying Random Forest parameters into ARCH, producing volatility forecasts with interpretable time-varying coefficients.
- **Application:** Applied to USD/GBP, the model traces how Brexit and COVID-19 each shifted baseline volatility versus shock persistence, separately identifying channels that GARCH would conflate.

WORKSHOPS AND TRAINING

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| – Audited LSE PhD macroeconomics courses (EC555 and EC556) | 2025 – 2026 |
| – Tools for Macroeconomists Summer School | Aug 2026 |

GRANTS AND AWARDS

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| – Invited speaker (alongside Prof. Wouter den Haan), Erasmus University Trust Fund delegation to LSE 2026 | |
| – Full MSc scholarship funding: <i>Studiefonds Robert van der Rest</i> , <i>Fundatie van de Vrijvrouw van Renswoude te 's-Gravenhage</i> , <i>Hendrik Muller's Vaderlandsch Fonds</i> , <i>Bekker-la Bastide-Fonds</i> . | 2025 |
| – Distinction, MSc Econometrics and Mathematical Economics, LSE | 2025 |
| – Honours, Double BSc in Econometrics & Economics, Erasmus University Rotterdam | 2024 |
| – Thesis ranked top five in cohort, Erasmus University Rotterdam | 2024 |

EXTRACURRICULARS

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| – Financial Implementation Consultant (part-time), <i>Cevinio</i> | 2021 – 2024 |
| – Mandarin Language Study (HSK1–HSK4): <i>Shenzhen University</i> , <i>Chinese University of Hong Kong</i> , and <i>Beijing Language and Culture University</i> | 2019 – 2024 |
| – Statistics Teacher, <i>AthenaStudies</i> (VU Amsterdam and Erasmus University) | 2021 – 2023 |
| – Treasurer, TokoTalks Committee, <i>A.S.C./A.V.S.V.</i> | 2021 – 2023 |
| – Varsity Rower, <i>Skadi Rotterdam</i> (heavyweight crew, Dutch national level) | 2019 – 2022 |

MISCELLANEOUS

Languages:	Dutch (native), Spanish (native), English (fluent), French (B2), Mandarin (B2)
Nationalities:	Dutch – Colombian – Canadian
Test scores:	GRE Quantitative 170/170, Verbal 160/170